

The Contract Year in the NBA: Possible Effects on Players Effort and Production

Abstract

The NBA features some of the highest paid players in all of sports, with \$50M+ AAV contracts occurring frequently and rising as time passes. A common question pondered by many sports fans is whether players will purposefully raise their performance and level of effort in their contract year (the year their current contract expires) in order to secure one of these lucrative contracts. To approach this area of speculation, this paper will explore whether a player being in their contract year has an effect on their play. Using data on as many players as possible from past NBA seasons, I will utilize regression techniques and a variety of statistics to ascertain possible connections.

The topic of the contract year and its impact has been explored previously in the literature, notably by Kevin Stiroh in 2007 and David Berri et al in 2006. Stiroh's paper, entitled *Playing for Keeps: Pay and Performance in the NBA* utilized data from the 1980s and 1990s on player performance in and directly after contract years. The paper indeed found a connection between greater performance in the contract year and decreased performance after a new contract, with performance being measured by a "Composite Performance Rating" composed of 6 basic statistics. The paper by Berri et al entitled *Shirking on the Court* used 2 different regressions and focused on the prevalence of "shirking", the act of applying less effort after a contract is signed. In the first regression, using a player efficiency composite statistic, evidence was indeed found for shirking in the NBA, however in the second regression using a "Wins Produced" statistic, the evidence was significantly diminished. This led the paper to conclude that players do not necessarily apply less effort after a new contract, but perhaps "stat pad" in contract years. These are all very insightful findings that this paper will incorporate in order to ascertain as clearly as possible the impact of the contract year.

Firstly, I will use multiple regressions to evaluate different measures of player performance: an advanced statistic measuring player performance (RAPTOR) and the level of the players effort on the court (hustle index). The hustle index is a composite metric developed by PerThirtySix that is composed of niche statistics such as box-outs, average running speed, and drawn charges. By using multiple measures of performances, greater insight can be achieved regarding a players true intentions; whether they are applying more effort, stat padding, or simply performing better. In order to build on the already established literature and provide an up to date analysis, I will focus on the most recent NBA seasons. Data will be sourced from Basketball Reference, which provides salary information as well as basic and advanced statistics, and hustle index statistics will be sourced from PerThirtySix, which provides a single statistic for each player every season. The remainder of this paper will provide an overview of the current literature, review of methodology, results of the regressions, and a discussion of their implications.

Literature Review

Labor Economics and the Sports Laboratory

The contract year phenomenon, at its core, stems from a direct application of foundational labor economics—particularly focusing on the issues of principal-agent problems, asymmetric information, and moral hazard. In traditional corporate environments, a principal (the employer or sports organization) often struggles to perfectly monitor or enforce the day-to-day output level of their agent (the employee or player). Because of this monitoring gap, agents inherently possess asymmetrical incentives to vary their effort based on short-term financial gains. Over the past few decades, economists have rigorously tested whether or not players deliberately adjust their on-court behavior when approaching the unrestricted free-agent market, thereby artificially inflating their statistics to secure larger, guaranteed deals, and subsequently "shirking" (adjusting performance and effort downwards) once a lucrative, multi-year deal is signed.

To understand why this phenomenon is so frequently studied within the realm of athletics, it is vital to look to Lawrence Kahn's (2000) foundational paper, *The Sports Business as a Labor Market Laboratory*. Kahn established that professional sports leagues offer economists a pristine, unparalleled testing ground for labor theories that are nearly impossible to measure in traditional corporate settings. In a standard office, quantifying an individual worker's exact daily marginal product is highly subjective. In professional sports, however, performance is meticulously recorded every single night through standardized box scores, and salaries are publicly available. Because basketball is a dynamic sport that heavily depends on individual talent and autonomous on-court decision-making—primarily due to how much of the sport's action is directly controlled by a single player possessing the ball—the NBA became the ultimate laboratory for testing strategic effort and moral hazard.

Evidence of Strategic Effort in the NBA

Because NBA contracts are fully guaranteed upon signing, the league presents a massive asymmetrical risk for the franchise. Kevin Stiroh's seminal 2007 article, *Playing for Keeps: Pay and Performance in the NBA*, attempted to empirically estimate whether players actively exploited this structure by increasing their statistics in the seasons immediately preceding their free agency. Stiroh's research utilized a vast array of player statistic data spanning the 1980s and 1990s. Crucially, he employed a fixed-effects econometric model that controlled for the innate talent levels of the individual player, as well as for the normal biological aging process, ensuring that natural athletic peaks were not mistaken for contract year motivation.

Stiroh indeed found robust empirical evidence that players performed statistically better when their contracts were expiring than they did immediately after signing a long-term deal. He utilized a "Composite Performance Rating"—which aggregated six basic player statistics into a single metric—and found statistically significant drops in this performance rating during players' post-contract years. His work provided concrete evidence for a contract year effect, demonstrating that players deliberately change their

level of output to maximize their returns upon contract renewal. The potential implication of this result for NBA front offices warrants caution: the assumed benefit of locking a player into a long-term, guaranteed contract is often clouded by performance manipulation, as the player's motivation becomes uncertain the moment financial security is achieved.

Stat Padding vs. Genuine Effort (Shirking)

However, a critical nuance regarding *how* players manipulate their performance was introduced in David Berri and Anthony Krautmann's 2006 paper entitled *Shirking on the Court*. Berri and Krautmann essentially recreated Stiroh's study, but they utilized two different dependent variables, ultimately concluding entirely different things depending on how player output was actively measured.

In the first regression that utilized a standard Composite Performance Rating, similar to Stiroh's calculation, they discovered highly consistent evidence of the traditional shirking phenomenon, demonstrated by decreased raw volumetric statistics of players post new deal agreement. However, in their second regression when utilizing "Wins Produced"-a metric that attempts to isolate a player's true on-court marginal contribution to team victories rather than rewarding volume in raw statistics-evidence of post-contract shirking dropped significantly.

This dichotomy leads Berri and Krautmann to an interesting conclusion, that, although the evidence may not suggest players expend less effort to win basketball games post contract deal, players may simply engage in "stat padding" pre-contract expires. "Stat padding" in the context of the NBA may occur when a player chooses to selfishly chase statistics that may be historically rewarding from a financial perspective within contract negotiations and free agency-such as demanding higher usage rate in order to score more points or gambling to take loose balls to accumulate more steals-in the process disregarding less obvious on-court fundamentals that would benefit his team more directly, such as setting harder screens or running through more defensive rotations.

Comparison of other Leagues: The MLB contract year

Understanding the contract year phenomenon requires setting the context for what incentives look like across different sports which operate within different constraints and reward structures. Major League Baseball (MLB) is an optimal sport for comparison because the high variance, individualized, discrete nature of the game presents a very different structure of opportunity than the fluidity of the NBA.

Working with a broad range of hitters from 2004 to 2008, Hummel and O'Neill (2014) conducted a fixed effects analysis of MLB free agents. They discovered a small, though positive 4.2% to 5.5% increase in players' OPS during their contract years. To provide context for the limited gain, Hummel and O'Neill cite a famous initial paper by Lehn (1982) where it was discovered that MLB players were significantly more likely to have taken a trip to the Disabled List after signing their long-term, guaranteed contracts. Lehn's results pointed to a reduction in effort but only with respect to a player's willingness to play through minor injuries rather than effort within the box when the financial contract was already agreed upon.

Compared to studies of NBA players, there seems to be less evidence of a fundamental change in on-court play in baseball research. This emphasizes an important aspect of sports econometrics-player interdependency. It is more difficult for a player to increase their rate of success in MLB given that they simply don't have enough opportunity to exert significant change to their game. A baseball player simply cannot choose to shoot more to increase scoring. While one could choose to swing more often at any pitch, doing so would negatively impact batting average and would hurt their bottom line during contract negotiation. A basketball player however, is able to force his team into using him more often, can dictate to teammates that he needs the ball, and generally assert more individual dominance over the flow of the game and thus have a more obvious statistical output that has historically driven better compensation in free agency.

Conclusion of Contract Year research

It can be confidently concluded, therefore, that the contract year is not a universal phenomenon and is highly dependent on the specific circumstances within a league or organization. While it can be reasonably assumed that professional athletes generally seek increases in compensation, the stat-padding phenomenon is entirely contingent on the ability of the individual player to exert their will upon the game and affect their individual outcomes.

It is therefore a player's level of individual dominance over their respective playstyle and its outputs which is responsible for the success of the contract year phenomenon within particular leagues like the NBA. As exemplified by the research of Kahn, Stiroh, and Berri et al. The NBA's use of guaranteed contracts and its reliance on individual statistical outcomes makes it the perfect playing field for the ongoing competition between true athletic ability and financial self-interest.

Data

TABLE 1
Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>RAPTOR</i>	422	0.73	2.47	-6.50	8.07
<i>Hustle Index</i>	422	51.49	19.82	4.39	115.20
<i>Contract Year</i>	422	0.25	0.43	0	1
<i>Opt-Out Status</i>	422	0.05	0.22	0	1
<i>Age</i>	422	26.06	4.02	19	40
<i>Age²</i>	422	695.23	220.76	361	1,600

A comprehensive panel dataset has been compiled to empirically assess the contract year's effect on player performance and effort, primarily using four separate data sources. Basketball Reference, PerThirtySix, Spotrac, and historical statistics archives were utilized to gather individual level player data. In an attempt to build off of the foundations laid by works such as Stiroh (2007) and Berri et al. (2006) where the sample sizes consist of data over multiple decades, a focused approach was taken to narrow down the sampling period to the modern era of NBA basketball; namely, the 2022-2023, 2023-2024

and 2024-2025 NBA seasons. In order to measure the individual player's trends over time and to eliminate individual level talent when implementing player fixed effects, a minimum of three seasons were needed for each individual. Any player whose contract year falls within this specific three-season window were then selected, and player-level data was gathered from each season that each selected player played in to form the player base for this project. After cleaning and merging data from each respective source, there was a total of 422 individual seasons played by roughly 140 different players in the sample.

The dependent variables were carefully chosen in order to be able to distinguish between general player success on the court and true physical effort. This distinction is necessary given the finding of Berri et al. (2006) that players may simply not expand more actual effort to win games during a contract year, but rather "stat-pad" with the traditional box score metrics. The performance-based dependent variable selected is therefore RAPTOR (Robust Algorithm using Player Tracking and On/Off Ratings), a relatively complex metric designed to measure a player's individual marginal contribution to their team. Through utilizing this comprehensive metric for my dependent variable (obtained from various online historical archives such as Neil Paine's substack), a sophisticated and comprehensive evaluation of individual player performance can be measured.

Since a performance metric is still inherently focused on outcomes and not pure exertion, a separate regression with a second dependent variable was included to more closely monitor player effort. This statistic, labeled the Hustle Index, uses highly granular on-court physical indicators that are virtually impossible to stat-pad without true physical effort, such as total distance ran, speed of running, box-outs, and charges taken. Statistics were gathered from PerThirtySix and then aggregated by specific weights dictated by the formulas outlined on their website. This final statistics were compiled and a fixed base number was added to avoid any negative numbers and scale all data between 0 and roughly 100. Importantly, both the Hustle Index and RAPTOR metric are irrespective of time, ensuring there is not a bias towards players who receive more playing time.

Specifically, RAPTOR is a plus-minus statistic measuring a player's net impact on the game, and the Hustle Index measures statistics per 36 minutes.

TABLE 2
Hustle Index: Components and Weights

Component	Weight
<i>Average Speed</i>	5
<i>Box Outs</i>	3
<i>Charges Drawn</i>	7
<i>Contested 2pt Shots</i>	3
<i>Contested 3pt Shots</i>	3
<i>Deflections</i>	7
<i>Distance Traveled</i>	5
<i>Loose Balls Recovered</i>	7
<i>Screen Assist Points</i>	3

The components used in the hustle index, all per 36 minutes, and their relative weights as they contribute to the final metric.

The primary independent variable of interest—the contract year status—required careful classification to accurately reflect the true economic incentives faced by the players.

Contract year statuses, as well as active contract structures and data, were sourced from Spotrac. All available player contracts were exported, and an XLOOKUP function was utilized within Excel to systematically mark each player-season observation with a binary dummy variable (1 if the season was a contract year, 0 otherwise).

The nature of modern, complex contract structures forced a refinement of this variable, as the ubiquity of the player option created a severe threat of endogeneity and reverse causality to the model. Theoretically, a player's output should be a function of the player's contract. When a player has a player option, the output of that player may become a determinant of their contract status. For example, if a player underperforms in a

season and has a player option that they decide to opt into out of financial security, their underperformance determines the status of the contract; if they perform poorly, the performance induces the status of the contract year. This is reverse causality. On the other hand, a player may have an excellent season and choose to opt out of their option in order to secure a more lucrative contract. In this case, their performance was the cause of them being in a contract year. To eliminate this reverse causality, any player who opted out of a player option was separated within the data by creating a separate status for them. All player options across the three seasons were retrieved from Sportrac and the variable "Opt-Out" was constructed using Excel XLOOKUP to see if the season being observed was one year before an opted-out player option was executed. If that indicator variable was 1, the default contract year dummy variable was automatically changed to 0. This way, the main contract year variable only measures natural, non-player driven contract expiration. Additionally, the dummy variable for opt-out status was included in the regression to observe any illuminating effects.

In addition to the main variables described above, control variables were added to help isolate the effect of the contract year. As is the standard in an examination of professional sports, the variables Age and Age-squared were included to capture normal, non-linear decline in physical performance and the expected pattern that players performance will increase and then decline with age irrespective of contract status. Additionally, a Current Team variable was added to account for effects that player systems, team coaching, and spacing can have on player RAPTOR and Hustle Index stats.

Although the dataset provides a robust base for elucidation of broad league trends, a critical weakness exists in the nature of this data, which is survivorship bias. Since the data can only come from available, current NBA contracts and since Sportrac is limited to listing the 100 most valuable contracts for each season, the sample set is limited to only highly valuable, employed NBA players. This means that players who potentially had a poor contract year, could not earn a new deal, and exited the league are non-existent in the data. This means a significant portion, specifically the lower end, of the performance

distribution has been greatly reduced; therefore, the results will somewhat overestimate the positive impacts of the contract year due to only the survivors being modeled.

Methods

To effectively test if players in the NBA change the intensity of play and effort exerted as their free agent status approaches, this paper adopts the framework of a panel data fixed effects regression. In examining the contract year phenomenon, there exists an inherent and ubiquitous econometric problem – unobserved heterogeneity. A straightforward cross-sectional comparison of players with and without contract years would suffer from considerable omitted variable bias. An athlete's inherent skills, natural work ethic, height and wingspan, and raw basketball intuition are clearly highly correlated with their contract situation (as players with the best underlying talent are afforded longer, more numerous contracts), and therefore will clearly correlate with their statistics. Comparing the statistics for an MVP level player in a non-contract year to those for a bench-warmer in a contract year would lead to severely biased and uninformative statistics that confuse inherent skill level with contingent effort.

To mitigate this structural problem, and following the methodology of foundational papers of the past, this paper relies on player fixed effects. Player fixed effects will effectively allow us to consider only intra-player variation and control for any inherent differences in player ability. We are able to establish for each individual an expected baseline level of play based on that player's historical numbers and control entirely for player fixed effects. Thus we are not comparing Player A to Player B, but Player A's statistics in his contract year to Player A's performance levels during his non-contract years.

The basic empirical specifications for this study take the form of the two regression equations below:

$$RAPTOR_{it} = \beta_0 + \beta_1 ContractYr_{it} + \beta_2 OptOut_{it} \\ + \beta_3 Age_{it} + \beta_4 Age_{it}^2 + \beta_5 Team_{it} + \alpha_i + \epsilon_{it}$$

$$HustleIndex_{it} = \beta_0 + \beta_1 ContractYr_{it} + \beta_2 OptOut_{it} \\ + \beta_3 Age_{it} + \beta_4 Age_{it}^2 + \beta_5 Team_{it} + \alpha_i + \epsilon_{it}$$

Regression equations, with i denoting individual player and t denoting a given NBA season. Player fixed effects are denoted with α_i .

The dual dependent variables of RAPTOR and HustleIndex were carefully chosen as to isolate and test the distinct aspects of the effort vs. performance problem described by Berri et al. (2006). While traditional box-score statistics such as points and rebounds can potentially be biased towards certain playstyles and overall capture less of a player's true performance, RAPTOR is a more sophisticated, modern composite metric, developed using player-tracking technology and on-court/off-court point differentials that estimates a players actual, marginal impact on the success of the team. RAPTOR cannot be directly manipulated in the same way as points; however, it is ultimately a metric for output rather than for effort.

It is for this reason that equation 2 includes the Hustle Index as its dependent variable, as this measures pure, unglamorous, physical output-the amount of distance run, number of defensive rebounds generated, number of charges taken, etc. The model runs regressions of both RAPTOR and HustleIndex against the control variables in order to differentiate actual effort from artificially inflated performance. A statistically significant β_1 in both models would imply that the contract year indeed has an effect on player effort and performance. An alternative is that β_1 is significant in the RAPTOR model but statistically insignificant in the HustleIndex model, and this would give statistically

robust evidence to Berri's stat-padding hypothesis. That is, players are attempting to boost their visible market value as a free agent by increasing their raw performance while keeping their effort level constant, essentially shifting their effort from supporting the team to boosting their individual statistics.

In both models, the coefficient of most interest is β_1 which measures the change in players' play resulting from the contract year dummy variable. The contract year dummy variable takes the value of 1 in any season when the player is in their final contract year on their standard NBA deal; otherwise, it is 0. If 1 is positive and significant, then it provides a direct empirical test of whether or not the contract year effect actually takes place in the NBA. Since NBA deals today are entirely guaranteed, a player experiences highly differing points on the motivation timeline, as a great deal of incentive exists to play above average and exert a greater effort right before becoming a UFA, thereby hoping to obtain a much richer new deal. However, after said deal is completed and all money guaranteed, a greater degree of urgency to maintain a state of optimal play no longer exists. Hence, β_1 measures exactly how much better on average a player performs when their long-term financial future is at stake.

The second independent variable that must be controlled for is the player option, labeled β_2 . As seen in the data methodology, this is where most of the selection bias is rooted. In a theoretically sound model, the date that the contract expires is exogenous. However, with the introduction of player options, the agent is free to actively decide whether the contract expires or not based on whether or not their current production warranted the decision. It is more probable that an injury-prone, poorly performing player will opt into the contract for financial security in which case their poor performance, not the year of the contract caused their contract status. The existence of the OptOut variable primarily serves to isolate these cases from our main independent variable, contract year status, and examine β_1 as it pertains to true contract years, not contract years influenced by player option decisions.

To try and further differentiate the contract year effect from natural career development, a vector of time-varying controls is employed. We include the conventional Age term as well as a quadratic term Age^2 (β_3 and β_4). These variables will capture natural increases in performance over a player's career while also incorporating a natural physical decline past a certain point (the player's "prime"). NBA players' careers are typically marked by an initial acceleration of development into their early-twenties, peaking in athletic and skill capacity between the ages of 26 and 28 and then gradually declining physically.

The last variable is a Team variable (β_5) used to control for environmental, systemic and coaching effects. The quality of a player's surroundings can be reasonably assumed to directly affect their potential to manipulate their own statistics and a player's RAPTOR or Hustle Index can be dramatically increased or decreased simply by switching teams. A player that moves from a defensively oriented, slow-paced team to an up-tempo offense that provides good floor spacing for three-pointers will naturally have an increase in certain metrics because of the better environment-not because of contract induced athletic improvement. The team control variable attempts to control for such effects.

The idiosyncratic error term, ϵ , is added to the end of both equations to control for any remaining unobserved time-varying shocks in player performance. This includes positive or negative shocks to shooting efficiency, or off-the-court problems (such as mid-season injuries) that could affect performance in a given year relative to their expected output.

Empirical Results

The main aim of the regression is to identify whether NBA players alter their play and physical output on a statistically significant level during the last year of their contracts. In order to ascertain these effects, two separate fixed-effect panel regressions were utilized: the first, looking at the overarching performance level with the advanced RAPTOR stat; and the second, only raw physical input via the PerThirtySix Hustle Index. By including player fixed-effects we are able to control for differences in the intrinsic talent level of

individual players. Thus, we should interpret the coefficients below as within-player differences rather than differences between player type.

The descriptive statistics are helpful context when interpreting these results. For the 422 player-seasons, average RAPTOR was 0.73, with a standard deviation of 2.47 and range of -6.50 to 8.07. The player-seasons showed much variance for impact, but overall were close to a league average impact in most cases. The Hustle Index shows more variance, partially by design, with a mean of 51.49, but a standard deviation of 19.82 and a range of 4.39-115.20. About 25% of the player-seasons were contract years compared to 5% being opt-out seasons; this must be considered when viewing the results of the opt-outs, though they are of particular interest due to their significance.

TABLE 3
Regression Estimates (Dependent Variables: RAPTOR, Hustle Index; t-Statistics in Parentheses)

Variable	Coefficients (RAPTOR)	Coefficients (Hustle Index)
<i>Constant</i>	-53.082** (-4.9)	89.583* (1.7)
<i>Age</i>	4.197** (5.2)	-1.614 (-0.4)
<i>Age²</i>	-0.079** (-5.1)	0.002 (0.0)
<i>Contract Year Status</i>	0.301 (1.4)	0.722 (0.7)
<i>Opt-Outs</i>	0.093 (0.2)	-4.513** (-2.0)
R ²	0.19	0.13
F-statistic	2.16**	1.25

*Significant at 10% level.

**Significant at 5% level.

Performance Impacts (RAPTOR Regression)

The results from the first regression (a player's total on-court contribution estimated through RAPTOR) show significant findings regarding the modern contract year effect. The coefficient of primary interest, Contract Year Status, was 0.301 and was not

significant at the 5 percent or 10 percent level, although it is positive. Substantively, this indicates that the true on-court value contribution, measured by RAPTOR, of an average player on a standard contract year only increased by approximately 0.301 relative to their non-contract year own performance (holding age, age-squared, opt-out, and team context constant). This impact is not very large-the standard deviation for RAPTOR for the players in the sample is 2.47.

The fact that the Contract Year Status coefficient is not statistically significant means that players in today's NBA do not, on average, have any significantly larger impact on team performance measured by advanced statistics like RAPTOR than they would if their contract was not up for renewal (unlike the findings from Stiroh [2007]). RAPTOR is a more comprehensive measure of contribution to winning than a standard box-score metric is; RAPTOR doesn't reward raw inputs (points, rebounds, etc.) as much as it values a player's overall effectiveness on the court. As such, a player must increase their value significantly to move the RAPTOR metric with these commonly emphasized basic statistics. So while players do stand to benefit from increased contract prospects, it is not the case that, as measured by RAPTOR, there is a significant on-court impact when these incentives are available.

The coefficients of the control variables are strong in the RAPTOR regression, providing confidence in the overall model. Age is positively and significantly associated with RAPTOR at the 5% level, as is expected: with a t-statistic of 5.2 and a coefficient of 4.197, age contributes meaningfully positively to RAPTOR. However, age squared is negatively and significantly associated with RAPTOR (at the 5% level; $t=-5.1$, $\text{coef}=-0.079$), indicating the expected concave relationship between age and player performance. Age improves up until the athletic prime, but declines steadily thereafter.

Similarly, the Opt-Out coefficient (0.093, $t=0.2$) is positive, as expected, but the impact is not statistically significant; there's no evidence that opt-out status has any meaningful effect on player performance via RAPTOR. This result adds further credence to the

methodological choice of separately accounting for contract years, since the performance outcomes are so varied.

Overall, the RAPTOR regression provides very little evidence that the presence of standard contract year incentives improves a player's on-court contribution through RAPTOR. The R-squared for the model is 0.19, meaning the explanatory variables capture 19% of the variation within players' individual RAPTOR measures. The F-statistic is 2.16 and significant at the 5% level, meaning the model as a whole is significant. However, the only statistically significant effect in the regression was the age effect.

Effort and Exertion Impacts (Hustle Index Regression)

Although RAPTOR measured the result of a player's performance as an overall measure of on-court value, the Hustle Index was intended to capture a distinct variable – raw physical exertion and effort based play. The Hustle Index regression is more crucial than the RAPTOR regression in that it directly addresses the separation that Berri et al. (2006) highlights between actual effort and feigned performance. In other words, if contract years were indeed related to increased player effort, this effort would surely manifest itself in both advanced metrics and in effort-based plays, such as boxing out, running the floor, drawing charges, etc. – difficult-to-fake plays, if not played by a player putting in extra effort.

The coefficient on Contract Year Status in the Hustle Index regression is 0.722, with a t-statistic of 0.7. Similar to the coefficient on Contract Year Status for the RAPTOR regression, the estimate is positive but not statistically significant. In practical terms, this implies that playing in a standard contract year decreases a player's Hustle Index by only 0.722 points, relative to the player's baseline. Since the standard deviation on the Hustle Index is 19.82, the predicted increase in hustle is insignificant. This indicates no significant increase in the player's physical effort during standard contract years.

This observation is crucial in arguing against the notion of player effort across the board due to contract incentives. The player who tries "harder" across the board due to contract year pressure should still, at minimum, increase effort-based behaviors in the Hustle Index. Because of the tiny coefficient and insignificance, this is not evident. The estimated effect seems to be mostly stable or related in some other capacity. A player who routinely sprints, boxes out, or sacrifices his body for loose balls is likely doing so consistently over contract situations; a player who doesn't will not suddenly begin doing so simply because he is playing his final contract year.

The crucial observation in the Hustle Index regression is the coefficient on Opt-Out Status. The coefficient on Opt-Out Status is -4.513, with a t-statistic of -2.0. As indicated in the table of significance, the 5% level of significance is achieved. This coefficient represents the one unambiguous player performance-based contract dynamic within this regression. Unlike the standard Contract Year Status coefficient, which appears insignificant and positive in both cases, Opt-Out status provides a large, negative, statistically significant link with effort-based performance.

The size of the Opt-Out Status coefficient is also noteworthy. A player is predicted to be 4.513 points lower in terms of Hustle Index than his average if he is playing the year prior to opting out. A decrease of 4.513 from the average of 51.49 is significant in terms of effort-based metrics, even given the relatively low t-statistic of the coefficient. The Opt-Out coefficient suggests, clearly over all else in this regression, that there is indeed a difference between a situation in which a player's contract naturally expires and a situation in which he is in control of when the contract expires.

The results of this study have many implications for explaining player behavior. Usually it is expected that players work harder when they have much to lose or gain in the future. What we found out about players who have the option to get out of their contract is different. This result is highly unexpected, but many possible explanations can be offered. Injury risk is one: players who have the option to leave their contract have a good reason to stay healthy and avoid injuries in preparation for their next contract. Secondly, players

may decrease their effort in the case that their future is already secured, as in they already expect to sign a contract next year or are already certain of their plans for the future, perhaps knowing that they will opt out before the season starts. Additionally, the opt out may be an effect of burnout related to a disharmony with team culture, environment, or other life factors involving the specific team they are on or contract they are engaged in. This case would cause the player to express less effort on the court and be more likely to opt out.

This study also provides insight into player and team dynamics. Rather than simply modulating the overall level of play, contract structures, incentives, and dynamics could have a more nuanced impact and affect performance in less straightforward ways. An expiring contract does not necessarily imply greater performance, as seen in the regression results. In this case a player's primary objectives may not be to increase performance but rather preserves themselves for an expected contract in the future and safeguard against injury.

The other coefficients in the Hustle Index regression are not statistically significant. Age has a coefficient of -1.614 and a t statistic of -0.4, and Age-squared has a coefficient of 0.002 and a t statistic of 0.0. In contrast with the RAPTOR model, the age curve does not explain a significant amount of variation in hustle. This may imply that effort-based statistics are less susceptible to the standard age curve. Although performance clearly does peak and then decline with age, hustle is potentially more related to less discrete factors such as the role a player fills, coach and team expectations, play style, physical health and safety concerns, and personal psychology than to age.

The Hustle Index regression has R-squared of 0.13 and an F-statistic of 1.25; its overall explanatory power is inferior to the RAPTOR model's, and that is not surprising. Hustle statistics are prone to greater subjectivity and game-to-game variability which would be impossible to adequately control for in a regression analysis (e.g. The demands of the opponent, minute allocation, coaching schemes, player health status and risk-taking

behavior, role changes). Nonetheless, the strong significant negative impact of Opt-Out status stands out despite this noise, and is one of the paper's key empirical findings.

Control variables and endogenous sources

Together, the controls and contract-related variables indicate that the regression framework is capturing distinctions among kinds of player behavior. The significant age and age-squared coefficients in the RAPTOR model confirms that overall player value clearly follows the expected player career-arc shape, so we need to include quadratic age controls to be certain that the model is not picking up natural improvement near major contract periods as contract year effects. If it weren't for these controls, improvement over a young player's early career could be mistaken for effort increase due to contract motivations near the first big free agency period.

It's also necessary to separate out standard Contract Year Status from Opt-Out Status, which the current framework successfully does. If Opt-Out seasons were counted under the general Contract Year dummy variable, then very different kinds of economically important decisions would be treated as one; the regression results show how significant this is. Coefficients for standard Contract Year are small and statistically insignificant in both regressions. Coefficients for Opt-Out season are neither statistically significant on RAPTOR nor on Hustle Index, but have a statistically significant negative impact on hustle. This significant impact would have been impossible to discover if we had not split out Opt-Out years from general contract years.

This separation helps remedy the problem of endogeneity discussed in the methods section. While standard contract years are generally predetermined by player contracts, the player partially controls whether he opts out of his contract and has substantial discretion over whether or not to exercise this choice based on his estimate of his own market value in free agency. Because opt-out decisions themselves partly contribute to this market value, opt-out seasons are somewhat strategically different than standard contract years. The negative Hustle Index effect of Opt-Out Status implies that it is

possible player effort levels do not differ on the basis of individual player traits, but rather because the benefit that comes from risk-taking does not sufficiently outweigh the potential for future injury and its impact on contract negotiations in an opt-out season.

Synthesis and Analytical Limitations

Overall, when taking the regression output as a whole, there is no compelling evidence of the classic contract-year effect in today's NBA. In both regressions, Contract Year Status is measured, though neither estimate is close to being statistically significant (coefficient 0.301 with t-statistic 1.4 in RAPTOR; coefficient 0.722 with t-statistic 0.7 in Hustle Index). Clearly neither estimate is large or precise enough to suggest that an ordinary expiring contract simply increases on-court true performance or raw physical output.

The only significant finding comes instead from the Opt-Out variable in the Hustle Index regression: a coefficient of -4.513 (statistically significant at the five percent level) indicates players with an opt-out on their contract take fewer physical risks. This is certainly not a moral hazard result; the player with the opt-out incentive responds differently than the player whose contract is up in two years. That is, avoiding injury may be more rewarding in these circumstances or the player may already be certain about their future in the league.

This may simply have to do with changed evaluation from the front office. In Stiroh and Berri's studies, the performance was based on simple composite metrics mostly based on traditional box score stats. Now, as discussed above, modern evaluations are far more sophisticated, rely more on the advanced analytics discussed above, track physical outcomes more closely, include medical information, and feature more precise role-specific scouting than was perhaps prevalent then. Players in this more refined evaluation system might not have the same incentives to simply boost stats, but there is no doubt about the huge value of today's contracts, which could further exacerbate the avoidance incentives for the player with an opt-out.

Last, none of the above should be read without the usual caveats on survivorship bias. Because the analysis used active Spotrac contract data, players whose contract-year stats fell below a minimum threshold and subsequently had trouble getting an NBA contract may have fallen out of the sample. As such, the weakest contract-year performances may be under-represented. If all players who performed below expectations in contract years were included, the strength of contract-year effects would likely be even smaller. Nevertheless, the modern contract-year effect may not be what most imagine (that players indeed try harder during a contract year), but it still exists. In fact, the most surprising result is not that contract-years do nothing to incentivize on-court output or even hustle play; the opposite of hustle play is incentivized in a particular case, demonstrating that different incentives created by contract structure matter.

Conclusion and Directions for Future Research

The goal of this paper was to empirically examine the existence, magnitude and changing dynamics of the contract year phenomenon in the current National Basketball Association. Building on the two core concepts from labor economics-the principal-agent problem and moral hazard-the theoretical prediction regarding professional sports was quite simple: that rational, utility-maximizing athletes may have incentives to change their effort and thus alter on-court play when their contract expires and they are due to become a restricted or unrestricted free agent and thus maximize the future value of the player contract. The unequal risk shared by the principal (team franchises) and the agent (player) is particularly critical in the NBA where all contracts are guaranteed, and average annual salaries continue to increase significantly over time. Updating the seminal work of Stiroh (2007) and Berri et al. (2006) to include current panel data of NBA play from the 2022-2023 to 2024-2025 seasons, this paper attempts to identify whether players still manipulate their output. More specifically, given the ability to model both the true impact of play (RAPTOR) and total on-court activity (Hustle Index) by using these as our dependent variables, the goal was to determine whether players increase genuine value-adding play, or just effort for the sake of seeming to do so.

Our fixed-effects regressions results actually paint a complex picture, quite contrary to the implications of the simple contract year story. In summary, the traditional contract year effect is weak and mostly insignificant in the modern NBA, less significant than in some historical studies. The Contract Year Status variable yielded a positive 0.301 in the RAPTOR regression but was insignificant because $t=1.4$. The coefficient on the Contract Year variable for the Hustle Index regression was also positive, a 0.722, but again statistically insignificant because $t=0.7$. These results indicate that a standard expiring contract does not necessarily lead to a league-wide statistically significant increase in true on-court impact or actual observed activity. Both coefficients did show an increase but were not significant, as they were quite small and ultimately not impactful in terms of the standard statistical benchmarks of significance.

The insignificance of the contract year effect, in particular, challenges the simpler moral hazard story in its most obvious form. If players were trying to actively and intentionally inflate their statistics and on-court impact then one would expect statistically significant results given player contracts in the NBA are extremely large. Thus, the standard theory that expiring contracts have a positive effect on player performance must be somewhat limited. Instead, perhaps, the contract year effect is highly inconsistent across players or is canceled out by other factors, such as changing roles, health issues, coaching styles, and natural aging curves. It does not necessarily imply that athletes are not influenced by future salary but that simply having an expiring contract is not enough to inspire an observable statistically significant change in on-court activity or actual impact.

The most compelling and interesting result of this paper, contrary to common assumptions, is neither from the standard Contract Year variable nor even the variables associated with RAPTOR, but from the Opt-Out Status variable in the Hustle Index regression. The Opt-Out Status had almost no relation to RAPTOR, yielding a coefficient of 0.093 with a t -statistic of 0.2, but it had an extremely significant and large effect on hustle: the Opt-Out Status coefficient in the Hustle Index regression was -4.513 and significant at the five-percent level, with a t -statistic of -2.0 . That is, on average, when

players had the chance to opt-out their year prior to the study's estimate of player Hustle Index were nearly 4.5 points lower than on any other point along their career arc. Since the average hustle player in this dataset has a Hustle Index score of 51.49, a decline of nearly 4.5 points is fairly substantial.

This result is not insignificant. The traditional contract year hypothesis states that players will increase their work ethic when future wages are at stake; however, the results from the opt-out model suggest that players will reduce some effort when the incentives that a new contract provides have a positive or negative element. A possible explanation of this phenomenon would seem to be risk-minimization. Players are making a strategic decision of when to re-enter free agency; the loss of value due to the risk of a season-ending injury and subsequently becoming untradeable is huge in this scenario and perhaps outweighs any incremental value added from a few hustle plays. Diving for loose balls, taking charges, pushing through physical screens, sprinting down the court, or constantly exerting high physical effort are all things that benefit the team, but they all significantly increase the risk of injury and fatigue. A player seeking to opt out may have a far more pressing concern to protect their body for a new, lucrative contract than that of any minor statistics improvement.

This is a modification of the principal-agent theory, not a negation of it. The problem is not necessarily that agents take advantage of their guaranteed contract; rather, it may be the nature of the contract incentives that provides different behavior stimuli. The contract that allows for opt-out incentivizes the agent to control whether they want to become a free agent, thus providing an additional layer of complex incentives. In these situations, it could be concluded that the agents may be working to lower downside risk rather than raise effort and stat production and this should thus be considered one of the most important contributions of this paper in that it highlights the possibility that the modern day contract year phenomenon may not be related to stat padding as proposed by Berri.

The results of the age controls further help to solidify the importance of adequately modelling player development. Both the age and age squared term in the RAPTOR

regression are statistically significant at the five percent level, and each has a positive and negative coefficient respectively (4.197 and -0.079). This implies that there exists a curvilinear progression in players' playing ability that aligns with typical physical development, with players increasing their efficiency and overall skill-sets with experience before declining with physical setbacks of old age. This becomes especially relevant when considering when players will want to achieve important contract moments-in many cases it will be around the height of their athleticism. Without including age and age squared, a contract year term might interpret an increase in a player's game as solely a consequence of their increased motivation to produce statistics while in an impending contract negotiation, rather than a natural increase in physical capacity. It is important to note that neither term proved statistically significant in the Hustle Index regression. This shows that raw physical exertion-oriented behavior is probably not tied as closely to the physical deterioration as it is to other aspects of a player's role, personality, risk tolerance, team expectations, and incentives presented by the contract itself.

Finally, the dampening of the modern contract year phenomenon can almost assuredly be traced to the pervasive adoption of analytics in the NBA. Whereas during Stroh's and Berri's eras, GMs relied primarily on raw traditional statistics (points, rebounds, etc.) that lent themselves to such statistical manipulation for team benefit (such as Berri describes), teams in this current era rely on advanced statistics that evaluate overall efficiency rather than quantity of performance in traditional box score statistics. Players who are willing to force shots or make marginal hustle plays, in order to slightly improve a traditional box score statistic, might be viewed more negatively on advanced metrics and as having negative WS or VORP values rather than positive. Thus, in this new era of information equality, there is less incentive to stat pad in the traditional way.

A second important reason for the muted results could be that all NBA players were grouped into a single model. The contract year effect is unlikely to be the same for all NBA players. It's likely that the performance/incentive effect is only strongly felt by mid-

career rotation and role players who may have their entire financial futures riding on the next contract that will hinge upon a single season's performance. It's probably unlikely that this effect is felt by superstars who are almost guaranteed maximum contracts if they stay healthy. It's also likely that fringe and minimum-salary players operate in a constant contract year state as their position on a roster is always in question. Grouping all players into one regression model may dilute the impact by averaging this strong effect with players that it does not strongly influence. The dataset is also inherently limited.

Survivorship bias is probably the most serious issue here. The fact that all of the data for the contract year information comes from Spotrac means that data for poor contract year performers who may not have been rewarded with a contract may be missing from this sample. This sample probably understates the strength of the contract year spike. A third limitation is that the sample size of the opt-out group is small (only about 5% of the observations); thus the relationship that exists for this group is not robust and needs further verification. These are clear areas for future research. Long-run panel data sets over the past decade, including players who played less than desired or who did not get contracts after contract years should be analyzed. Players should be broken up by salary, role (superstar, starter, bench player, fringe player) and possibly by age group or contract status (free agency, arbitration etc.) and individual regressions run. Of course, opt-outs need to be analyzed in a much more rigorous and structured way. The negative relationship that exists between Opt-Out Status and Hustle Index is very interesting and shows that players may be opting-out of contracts in a way that impacts their game.

While NBA finances may never have been stronger, general manager acumen and the inherent competitiveness of the league have tempered the classic contract year spike. While players certainly desire increased compensation for performing well, generating this jump in on-court performance seems increasingly difficult with sophisticated evaluation. The positive result associated with opt-outs signals that the structure of a contract is very important and contract control by players may indicate increased cautiousness as it gives them leverage not to play the same style of game. With all of this

considered, it does not seem to translate into better performance with the traditional contract year motivation.

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